AITOR BRACHO

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DATA ANALYST

Analytics | Clean Energy | Data-Driven Project Management

TECHNICAL EXPERIENCE

RESEARCH ASSISTANT — August 2019 – October 2025

Duke University, Triangle Universities Nuclear Laboratory | Durham, NC

- Addressed gaps in nuclear databases by producing the first measurements of key nuclear waste isotopes, supporting nuclear energy development and security.
- Managed experimental projects, developing and overseeing unique experimental hardware to guarantee data integrity.
- Developed novel quantitative analysis frameworks for low signal, time series data, improving the accuracy and reliability of nuclear measurements.
- Led a team across multiple institutions, managing project timelines and schedules ensuring successful experimental completion while contributing to operations and data reporting as a key collaborator.
- Communicated experimental plans, results, and analyses to collaborating members and stakeholders through technical notes, oral presentations, and data visualization.

EXPERIMENTAL METHODS INTERN — June 2023 – September 2023

TerraPower, LLC | Bellevue, WA

- Generated quantitative analyses and built models to evaluate reactor safety in a next generation Molten Salt Reactor design.
- Collaborated with reactor safety engineers to define client requirements and identify key parameters for novel reactor power models.
- Used Python, industry standard libraries to create predictive models of reactor behavior for the Molten Chloride Reactor Experiment, supporting project design and safety evaluations.
- Directed research on reactor power fluctuations and collaborated with experts to develop actionable frameworks for reactor safety studies.
- Authored technical reports presenting findings and strategies to collaborating stakeholders.

POLICY EXPERIENCE

ENERGY STORAGE SYSTEMS FELLOW — May 2024 – August 2024

North Carolina League of Conservation Voters | Releigh, NC

- Produced actionable energy efficiency and cost assessments of Duke Energy’s Carbon Plan/Integrated Resource Plan to support energy policy and planning in North Carolina.
- Engaged with experts and external advocacy groups to evaluate energy storage practices and policy impacts across states.
- Analyzed federal and state policies, as well as national energy data, to benchmark and visualize North Carolina’s energy storage and renewable development.
- Authored reports on nuclear energy and energy storage providing data-driven recommendations to guide NCLCV’s policy positions and regulatory interventions.

EDUCATION

PH.D. | Nuclear and Particle Physics | Duke University — August 2019 – October 2025

BACHELOR OF SCIENCE | Physics | Florida International University — August 2016 – May 2019

SKILLS

Software Skills: Python, ROOT, RooFit, C++, Pandas, Excel: Used for quantitative analyses in nuclear and particle physics research, industry, and energy development applications.

L**anguages**: English (Fluent), Spanish (Fluent)